

LED TRAFFIC LIGHT ELECTRONICS

Comprehensive Hands-On STEM Curriculum Guide

Welcome to the complete educator and student implementation curriculum guide. This manual accompanies the **Planrs Academy Robotics Series**, specifically designed to introduce early-stage student innovators to fundamental electronic hardware, breadboard circuit routing, and essential rules of road safety through interactive, kinetic learning frameworks.

Course Module:	Robotics & Hardware Foundations (Level 1)
Target Level:	Primary / Grade 1 - 4 (Ages 6 - 10)
Core Focus:	Circuit Loops, Parallel Formations, LED Polarization, Traffic Safety
Duration:	90 Minutes (Total Session Runtime)
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1. Session Overview & Pedagogical Goals

The **LED Traffic Light** framework uses guided play to unpack active engineering concepts. By transforming abstract concepts like electric polarization, circuit pathways, and energy regulation into physical realities on a solderless breadboard, kids gain an intuitive spatial baseline for electronics.

Core Learning Outcomes

- **Hardware Literacy:** Students will identify and handle solderless breadboards, light-emitting diodes (LEDs), jumper wires, and dedicated low-voltage DC power modules.
- **Circuit Directionality:** Understand that energy flows from a source, travels through a specific positive-to-negative orientation (+ / -), and must return full circle to complete a circuit loop.
- **Civic Safety Mapping:** Reinforce physical roadway rules by directly pairing the physical state of automated electronic components to real-world pedestrian and auto laws.

Today's Fun Plan Structure

Every step of this module is structured to match the visual sequencing within the core slide pack:

1. **Meet Chintu:** Contextual immersion through our signature character's traffic jam dilemma.
2. **Find Our Parts:** Tactical verification and organization of individual hardware kits.
3. **Build It:** Step-by-step parallel wiring sequence execution.
4. **Play & Celebrate:** Kinesthetic gameplay loop (Simon Says) designed to audit performance and build confidence.

2. Storytelling Phase: Chintu in the Market Jam

Before unpacking hardware components, students must understand the real-world problem. This session initiates with an interactive storytelling format led by the instructor.

"Meet Chintu! Chintu is a cheerful yellow auto-rickshaw who loves driving through the bustling town market to deliver groceries! But today, the market intersection is an absolute mess. Zooming motorbikes, heavy pull-carts, and busy street shoppers are crowding from all sides! Honk-honk! Beep-beep! Nobody knows when to cross, when to yield, or when to move forward safely.

Suddenly, Chintu spots the issue: the intersection's primary traffic signal box is completely broken and unlit! The city needs smart inventors right away. Can YOU jump into the designer seat, build a new functional electronic signal box, and help Chintu solve the giant traffic jam?"

The Roadside Rules Matrix

Ask the classroom to repeat the core light operational meanings out loud while holding up corresponding color cards or markers:

Signal State	Operational Rule	Driver Behavioral Action
RED	Stop and Wait Safely	Brakes are locked, vehicles halt entirely behind the crosswalk line.
YELLOW	Be Ready!	Engines transition, speed drops, prepares to yield or safely match transitions.
GREEN	Go Safely and Smile!	Accelerate forward smoothly with active attention to intersecting paths.

3. Hardware Kit Inventory Analysis

Instruct students to unbox their custom Planrs Academy kit layout trays. Before starting electrical construction, verify that every individual possesses the exactly matched structural quantities detailed below.

Hardware Asset Name	Quantity	Visual Identification & Educational Functionality
Magic Board (Breadboard)	1 Unit	The core prototyping grid surface where electronic legs slot smoothly without glue or mess. Relies on internal hidden metal clips to bridge parts together.
Red Signal LED	1 Unit	Light-emitting diode encased in translucent red resin; standard 5mm node shell diameter.
Yellow Signal LED	1 Unit	Light-emitting diode encased in yellow resin; handles the intermediate transitional state.
Green Signal LED	1 Unit	Light-emitting diode encased in green resin; completes the standard vertical sequence array.
Low-Voltage Power Module	1 Unit	Dedicated hardware power base that links directly to the breadboard edge rails to feed energy path tracks.
Solid Jumper Control Wires	1 Set	Insulated multi-colored connections used to safely transfer control signals across isolated track blocks.

Instructor Safety Note

All elements in this student hardware kit operate beneath 5 Volts DC. The circuit paths present zero risk of electrical shocks. Remind students never to substitute custom household batteries or plug loose wire strips directly into standard mains outlets.

4. Anatomy of an LED: Understanding Electronic Polarization

A Light Emitting Diode (LED) behaves like a one-way street for electric energy. If energy approaches from the incorrect direction, the path remains firmly locked, and the node will not illuminate.

Identifying the Pins

Instruct students to hold an LED up at eye level. They will immediately notice that the two metallic lead legs are cut to distinct structural lengths:

- **The Long Leg (The Happy Leg):** Known formally as the **Anode**. This leg represents the positive terminal (+). It acts as the gate where energy must enter the component body.
- **The Short Leg (The Sad Leg):** Known formally as the **Cathode**. This leg represents the negative terminal (-). It acts as the exit where used energy flows out toward the return track.

The Flow Equation Principle

For illumination to occur across any diode path, energy dynamics must strictly balance:

$Energy_{\{In\}} \rightarrow Long\ Leg\ (+) \rightarrow LED\ Glow \rightarrow Short\ Leg\ (-) \rightarrow Energy_{\{Out\}}$

If an LED is accidentally placed backward, no damage occurs, but the loop is broken. Simply pull it out, rotate it 180 degrees, and re-insert it into the target column tracks.

5. Step 1: Mounting & Powering the Magic Board

The solderless breadboard is divided into specific target layout regions. The long horizontal columns tracking along the top and bottom edges are dedicated **Power Rails**, while the shorter vertical groups in the middle are component interaction columns.

ASSEMBLY DIRECTIVES

1. Take your rectangular **Magic Board** and orient it horizontally so the alphabetical grid labels run left-to-right.
2. Locate the **Power Module Base** from your component storage tray.
3. Align the dual structural underside pins of the Power Module exactly with the far edge holes of the breadboard tracking grid.
4. Gently apply downward vertical thumb pressure until the power base securely clicks flat into the breadboard face. Do not force or twist the component.
5. **Stop! Checklist Action:** Raise your hand for the lead teacher to inspect the structural link and safety-test the base before clicking any active batteries into place.

Under the Hood: How the Prototyping Tracks Run

Inside the plastic shell of the Magic Board sit rows of metallic spring clips. When you push a wire or component pin into a grid hole, these internal clips hold it snugly. Because these metal tracks run continuously behind the scenes, any components plugged into the same electrical path are automatically linked together—no solder or messy glue needed!

6. Step 2: Routing the Primary Red Signal Light

Now we'll insert the very first signal phase of our functional automated street junction box: the top Red LED indicator node.

ASSEMBLY DIRECTIVES

1. Locate the **Red LED node** from your unboxed tracking workspace tray.
2. Examine the pins closely to isolate the **Long Leg (+)** from the **Short Leg (-)**.
3. Locate **Row 5** and **Row 6** on the middle sector grid of your Magic Board.
4. Carefully push the **Long Leg (+)** into the terminal node tracking hole sitting inside **Row 5**.
5. Simultaneously push the **Short Leg (-)** into the adjacent terminal tracking hole sitting inside **Row 6**.
6. Ensure the plastic base of the LED floats cleanly roughly 5mm above the surface of the breadboard grid floor. Do not bend the leads flat against each other.

Active Circuit Loop Status Verification

Target Pin Check	Assigned Board Destination	Expected Functional State
Long Pin Lead (Anode)	Row 5 Grid Intersection	Stands ready to accept positive incoming control voltage.
Short Pin Lead (Cathode)	Row 6 Grid Intersection	Connects downstream to facilitate return circuit loops.

7. Step 3: Expanding the Parallel Array (Yellow & Green Signal Elements)

To build a true traffic light array, all three color nodes must sit side-by-side in parallel tracking lines, each keeping its own isolated control pathway on the Magic Board grid.

ASSEMBLY DIRECTIVES

Part A: Yellow Phase Integration

1. Pick up the **Yellow LED node** and identify its long and short leads.
2. Find **Row 7** and **Row 8** on the breadboard layout field.
3. Insert the **Long Leg (+)** securely into the hole track of **Row 7**.
4. Insert the **Short Leg (–)** securely into the hole track of **Row 8**.

Part B: Green Phase Integration

1. Pick up the final **Green LED node** and map its corresponding leg polarities.
2. Find **Row 9** and **Row 10** on the breadboard layout field.
3. Insert the **Long Leg (+)** securely into the hole track of **Row 9**.
4. Insert the **Short Leg (–)** securely into the hole track of **Row 10**.

Visual Calibration Validation

Look down at the board from above. Your three LEDs should stand in a neat line, tracking across columns 5 through 10. Double check that all three long legs sit uniformly on the odd-numbered tracks (5, 7, 9) and all three short legs sit uniformly on the even-numbered tracks (6, 8, 10).

8. Step 4: Connecting the Jumper Routing System

With your signal LEDs and power base firmly in place, the final step is to hook up your multi-colored jumper wires. These wires act as switchable pipelines, letting us send energy precisely to each light node one-by-one.

ASSEMBLY DIRECTIVES

1. Take a **solid jumper control wire** (preferably color-matched red) and insert one end into the main positive edge power rail distribution node.
2. Plug the opposite end of this jumper wire directly into **Row 5**, right alongside the long leg of your Red LED.
3. Watch closely: the Red LED should instantly illuminate!
4. To change the signal phase, carefully pull the jumper wire out from Row 5 and move it into **Row 7** (the Yellow input track). The Red light will turn off, and the Yellow light will instantly glow!
5. Move the wire once more into **Row 9** (the Green input track). Watch the system transition smoothly to the green phase, just like a real street light!

Troubleshooting Logic Diagnostics Block

If any node fails to glow bright upon jumping the target row, trace these debug steps:

- ☐ Check the polarization layout: Is the long pin lead accidentally sitting in the even row? If yes, reverse it.
- ☐ Verify structural seating: Are the pins pushed down deep enough to touch the internal metal clips? Give them a gentle extra push.
- ☐ Validate line power status: Is the master switch on the power base toggled to the 'ON' position? Ensure the power base indicator light is active.

9. Classroom Interactive Game: Traffic Light Simon Says

Now that our physical prototyping builds are complete and fully operational, we will shift into our kinesthetic classroom validation gameplay loop. This session anchors real-world safety laws to swift physical reflexes.

Game Rules Formulation

- The lead class instructor takes the position of Chief Traffic Controller, standing at the front of the room.
- All students stand beside their project desks, ensuring they have plenty of safe physical clearance space around them.
- The instructor changes the working state of the automated demo junction signal box on the projector or calls out a target phase command loud and clear.

Phase Command 1: Red Signal Activated

Action Directive: Every student must instantly freeze completely in place like a stone statue. No moving, no laughing, no talking! Brakes are locked!

Phase Command 2: Yellow Signal Activated

Action Directive: Drop down into a half-squat driving posture, gently revving imaginary vehicle engines while marching in place softly. Be ready!

Phase Command 3: Green Signal Activated

Action Directive: Safe movement allowed! Walk rapidly in place or follow a designated circle path layout while maintaining a bright smile. Drive safely!

Elimination Context

If a student moves during a Red phase, or fails to transition when a new color activates, they step out for one round to act as an Assistant Safety Deputy, helping the teacher spot tracking errors across the rest of the active field.

10. Knowledge Audit & Structural Review

Student Name: _____ Date: _____

Exercise A: Connect the Tracks

Draw a clean solid pencil line matching every identified component part on the left to its correct hardware operational rule or row location on the right side:

Component Part Asset

1. Long Pin Lead (+)
2. Short Pin Lead (–)
3. Magic Board Grid
4. Red Input Routing Hole
5. Yellow Input Routing Hole



Assigned Target Target / Rule

- A. Row Destination Track 7
- B. Solderless Prototyping Breadboard
- C. The Positive Anode (Happy Leg)
- D. Row Destination Track 5
- E. The Negative Cathode (Sad Leg)

Exercise B: Short Concept Explanations

1. What happens if you flip an LED around and plug its short leg into a positive track? Does it break, or does it just stay unlit? Explain why in your own words.

2. Why do we need traffic signals at busy market junctions like the one Chintu got stuck in? How do these colored lights help keep people and vehicles safe?

11. Creative Design Canvas Zone

Mission Challenge: Draw your very own custom traffic signal intersection scene below! Color your active lights using bright Red, Yellow, and Green markers, and be sure to sketch Chintu the Auto driving safely through a perfectly organized market crossing.



Planrs Academy STEM Curriculum Lab Portfolio Activity Record Sheet